Inequalities.

For positive a, b, c, d prove the following inequalities

1. $a^2 + b^2 + c^2 \geq ab + bc + ca$.
2. $a^4 + b^4 + c^4 + d^4 \geq 4abcd$.
3. $a(1 + b) + b(1 + c) + c(1 + a) \geq 6\sqrt{abc}$.
4. $(a + b)(b + c)(a + c) \geq 8abc$.
5. $(1 + ab)^2 + (1 + cd)^2 + (ac)^2 + (bd)^2 \geq 1$.
6. Prove inequality $\frac{a^2}{4} + b^2 + c^2 \geq ab - ac + 2bc$.
7. For any x, y, z prove the following:
$$x^4 + y^4 + z^2 + 1 \geq x(xy^2 - x + z + 1).$$
8. For $n \geq 3$ prove inequality:
$$\frac{1}{n+1} + \frac{1}{n+2} + ... + \frac{1}{2n} > \frac{3}{5}.$$
9. Which of the numbers $2^{2^{\cdot^{\cdot^{\cdot^{2}}}}}$ (10 times 2) or $3^{3^{\cdot^{\cdot^{\cdot^{3}}}}}$ (9 times 3) is larger?
10. For $n \geq 1$ prove: $3^n > n \cdot 2^n$.
11. For $n \geq 4$ prove: $2^n < n!$.
12. Let $a, b, c \geq 0$. Prove: $ab + bc + ca \geq a\sqrt{bc} + b\sqrt{ac} + c\sqrt{ab}$.
13. Let a, b, c — be positive. Prove $\frac{bc}{a} + \frac{ac}{b} + \frac{ab}{c} \geq (a + b + c)$
14. n — natural number. Prove: $\sqrt{n+1} - \sqrt{n-1} > \frac{1}{\sqrt{n}}$.